Kirti Rai

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Professional Summary

Senior Backend Engineer with 3+ years at HSBC designing distributed systems in Java, Spring Boot, and Apache Kafka processing **5M+ daily transactions**. Proven record of driving **30%+ P99 latency reduction**, zero-downtime deployments, and leading end-to-end system design across critical financial pipelines. **INSPIRE Scholar** (DST, Govt. of India) & **L'Oréal India For Young Women in Science Scholar**.

Technical Skills

Languages	Java (Advanced), SQL, Bash
Frameworks	Spring Boot, Spring MVC, Spring Security, Hibernate/JPA, Maven
Messaging & DevOps	Apache Kafka (Producer/Consumer, Event-Driven Architecture), Jenkins CI/CD, Docker, GitHub
Architecture & Practices	Distributed Systems, Microservices, RESTful APIs, Event-Driven Design, Performance Tuning, TDD, Clean Architecture

Experience

HSBC — Pune, Maharashtra, India

Senior Software Engineer — *July 2025 – Present*

- Architect enterprise Java/Spring Boot services for a high-availability payments platform processing **5M+ transactions/day** at 99.9%+ uptime; designed Kafka event-driven microservices improving end-to-end throughput by **~35%** and fully decoupling asynchronous workflows across 4+ critical business pipelines.
- Led system design reviews across a team of 6 engineers, enforcing clean architecture and code-quality standards that reduced post-release defects by **~40%**; profiled bottlenecks, refactored hot paths, and tuned DB queries to achieve **30%+ P99 latency reduction** with **zero unplanned downtime** over two consecutive quarters.
- Drove root-cause analysis on production incidents across high-value transaction pipelines, improving system resilience, SLA adherence, and contributing to fault-tolerant architecture patterns adopted team-wide.

Software Engineer — *February 2024 – July 2025*

- Delivered **8+ backend features** end-to-end in Agile sprints using Spring Boot and Kafka; revamped Jenkins CI/CD pipelines and introduced structured logging, reducing MTTR by **~25%** with zero critical post-release bugs across all releases.
- Conducted **50+ code reviews** coaching junior engineers on SOLID principles, fault-tolerance patterns, and microservices best practices, measurably raising overall team code quality.

Trainee Software Engineer — *August 2023 – February 2024*

- Contributed production-ready backend features within the **first 3 weeks** of onboarding; deepened expertise in HSBC's large-scale Kafka and Spring Boot architecture through direct collaboration with senior engineers on critical payment service components.

Cognizant — Noida, India (Hybrid/Remote)

Programmer Analyst & CIM Cloud Developer — *November 2022 – July 2023*

- Built and maintained enterprise Java applications end-to-end across full SDLC — requirements analysis, development, QA, and deployment — supporting **3 concurrent client projects**; simultaneously gained hands-on cloud deployment and integration experience as CIM Cloud Developer on distributed infrastructure.

Projects

NiDar – Women Safety Application | *Published Research* | VIEW*, 2023* — 2022 – 2023

- Designed and built a production-grade real-time women's safety mobile application featuring SOS emergency alerts, live GPS location tracking, and automated multi-contact notification; engineered the full backend pipeline including REST APIs, real-time event processing, and notification infrastructure — demonstrating end-to-end ownership from system architecture to deployment.
- Authored and published a **peer-reviewed research paper** documenting system architecture, scalability design decisions, and real-world testing outcomes, showcasing research depth alongside hands-on engineering delivery.

Education

Shri Ramswaroop Memorial College of Engineering and Management — Lucknow, India

Bachelor of Technology – Information Technology — *2019 – 2023*

- **INSPIRE Scholar** – Department of Science & Technology, Govt. of India — merit-based national scholarship for top science achievers. **L'Oréal India For Young Women in Science Scholar** — awarded for outstanding academic excellence and demonstrated leadership in STEM.

Certifications & Training

Spring Boot · Spring MVC · Maven · IAB Digital Marketing & Media Foundations